

```

118         ei_printf(" %s: ", ei_classifier_inferencing_categories[i]);
119         ei_printf("%.5f\r\n", result.classification[i].value);
120     }
121     #endif
122
123     // Print anomaly result (if it exists)
124     #if EI_CLASSIFIER_HAS_ANOMALY
125         ei_printf("Anomaly prediction: %.3f\r\n", result.anomaly);
126     #endif
127
128     #if EI_CLASSIFIER_HAS_VISUAL_ANOMALY
129         ei_printf("Visual anomalies:\r\n");
130         for (uint32_t i = 0; i < result.visual_ad_count; i++) {
131             ei_impulse_result_bounding_box_t bb = result.visual_ad_grid_cells[i];
132             if (bb.value == 0) {
133                 continue;
134             }

```

Fig. 7: Code snippet for anomaly detection using the trained ML model

```

Last login: Tue Sep 30 11:16:38 on console
ashwini@Ashwinis-MacBook-Air ~ % edge-impulse-data-forwarder --clean
Edge Impulse data forwarder v1.24.0
(node:11092) [DEP0040] DeprecationWarning: The 'punycode' module is deprecated. Please use a userland
(Use 'node --trace-deprecation ...' to show where the warning was created)
? What is your user name or e-mail address (edgeimpulse.com)? AshwiniSinhaEFY
? What is your password? [hidden]
WARN Password is incorrect

? What is your user name or e-mail address (edgeimpulse.com)? ashwini.sinha@efy.
in
? What is your password? [hidden]
Endpoints:
  Websocket: wss://remote-mgmt.edgeimpulse.com
  API:       https://studio.edgeimpulse.com
  Ingestion: https://ingestion.edgeimpulse.com

[SER] Connecting to /dev/tty.usbmodem01
[SER] Serial is connected (0)
[WS ] Connecting to wss://remote-mgmt.edgeimpulse.com
[WS ] Connected to wss://remote-mgmt.edgeimpulse.com

? To which project do you want to connect this device? Ashwini / Health_Anomaly
[SER] Detecting data frequency...
[SER] Detected data frequency: 14Hz
? 2 sensor axes detected (example values: [0.68,0.96]). What do you want to call them? Separate the nam
? What name do you want to give this device? IndusHealth
[WS ] Device "IndusHealth" is now connected to project "Health_Anomaly". To connect to another project,
[WS ] Go to https://studio.edgeimpulse.com/studio/791007/acquisition/training to build your machine lea

```

Fig. 8: Connecting the device to Edge Impulse again for testing the ML model

Place a finger on the sensor to begin readings. Then, in the Edge Impulse interface, open the Data Acquisition tab and collect data under normal health conditions. Use the same label name for all data under normal health conditions (refer to Fig. 3).

After collecting sufficient data, the impulse needs to be designed. Go to the Impulse Design option and select the method for training the ML model. For instance, the model can be trained to classify patterns in values, detect abnormalities, or predict upcoming values based on trends. In this device, the model is trained to detect abnormalities in a person's health vitals, so anomaly detection is selected as the impulse design.

Next, features are generated

from the collected values. Once the features are generated, the corresponding parameters must be saved. Then, proceed to the next tab to train the ML model for anomaly detection. A list of features and axes will be displayed. Select a few features for heart rate and a few for SpO₂ that fall within a healthy person's range and pattern, and then proceed to train the ML model.

Deploying the ML model

After the ML model has been trained, go to the Deployment section. Select Arduino Library and click Build. This will generate the ML model for Arduino and allow the library to be exported (refer to Fig. 6).

Next, open the Arduino IDE and add the exported library. Modify

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the ESP32 sensor example code included in the library to integrate the MAX30100 sensor. Additional functions, such as displaying data or sending alerts over Wi-Fi or via a buzzer, can be added for anomaly detection. Finally, upload the code to the IndusBoard Coin.

Testing

Once the code is uploaded and the components are assembled on a breadboard (see Fig. 1), power on the device. If configured, real-time health vitals appear on the OLED display. Otherwise, the Serial Monitor can be used to view anomaly detection results when a finger is placed on the sensor.

Next, place the finger of a person with a heart condition on the sensor. The device will display an anomaly, whereas for a healthy person, it outputs 1. If an alert function such as a buzzer has been added in the code, it triggers accordingly. The results can also be shown on the OLED display.

The ML model can also be tested without using the ML model library. Upload the earlier data forwarder code, then go to Live Classification. Open the terminal window on macOS or Linux (or CMD for Windows) and run the command:

```
edge-impulse-data-forwarder
```

It automatically connects to the last device used. Then, open the Edge Impulse workspace and select Live Classification. Place a finger on the sensor to view real-time anomaly data directly on the Edge Impulse platform (see Fig. 8). **EFY**

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